

Note: happy to provide pdf of the actual paper if there's any you can't seem to access.

Code Editors

Basic functionality of a code editor (which we can adapt for a writing context):

https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Environment_setup/Code_editors

Intelligent Tutoring Systems (ITS)

What it is: https://en.wikipedia.org/wiki/Intelligent_tutoring_system#

W-Pal, a writing ITS for children. See this demo video, from 1:13:

<https://youtu.be/5hGNvxnoww8?si=kYgRB5F8WV-Cdssp>

This paper on literacy ITS (focus on “Writing-Focused Technologies”):

<https://eric.ed.gov/?id=ED577131>

This article on building ITS (see Bayesian Knowledge Tracing):

https://medium.com/@ashrivastava_40082/building-an-intelligent-tutoring-system-from-scratch-rt-bkt-83693fb21286

Maybe some of the papers at this link would be helpful:

<https://soletlab.asu.edu/intelligent-tutoring-systems-publications/>

This paper: <https://cdn.aaai.org/ocs/5895/5895-29774-1-PB.pdf>

Relevant papers:

A. Alam, “Harnessing the Power of AI to Create Intelligent Tutoring Systems for Enhanced Classroom Experience and Improved Learning Outcomes,” in Intelligent Communication Technologies and Virtual Mobile Networks, G. Rajakumar, K.-L. Du, and Á. Rocha, Eds., Singapore: Springer Nature, 2023, pp. 571–591. doi: 10.1007/978-981-99-1767-9_42.

(Old, but I thought was a cool ITS example with some similarities to what we’re doing) A. Keleş, R. Ocak, A. Keleş, and A. Gülcü, “ZOSMAT: Web-based intelligent tutoring system for teaching–learning process,” Expert Systems with Applications, vol. 36, no. 2, Part 1, pp. 1229–1239, Mar. 2009, doi: 10.1016/j.eswa.2007.11.064.

M. E. Jacovina and D. S. McNamara, “Intelligent Tutoring Systems for Literacy: Existing Technologies and Continuing Challenges,” 2017. Accessed: Sep. 11, 2026. [Online]. Available: <https://eric.ed.gov/?id=ED577131>

Multi-Armed Bandits Problem

What it is: https://en.wikipedia.org/wiki/Multi-armed_bandit

A better description of what it is, and algorithms to solve it:

<https://towardsdatascience.com/the-multi-armed-bandit-problem-a-beginner-friendly-guide-2293ce7d8da8/>

Yet another description of what it is:

<https://people.stfx.ca/jdelamer/courses/csci-531/lectures/multi-armed-bandit/the-problem.html>

Relevant papers:

Clement, B., Roy, D., Oudeyer, P.-Y., & Lopes, M. (2015). Multi-Armed Bandits for Intelligent Tutoring Systems (arXiv:1310.3174). arXiv. <https://doi.org/10.48550/arXiv.1310.3174>

Frenoy, R., Soullard, Y., Thouvenin, I., & Gapenne, O. (2016). Adaptive Training Environment without Prior Knowledge: Modeling Feedback Selection as a Multi-armed Bandit Problem. Proceedings of the 2016 Conference on User Modeling Adaptation and Personalization, UMAP '16, 131–139. <https://doi.org/10.1145/2930238.2930256>

Yanez, F. J., Zavaleta-Bernuy, A., Han, Z., Liut, M., Rafferty, A., & Williams, J. J. (2022). Increasing Students' Engagement to Reminder Emails Through Multi-Armed Bandits (arXiv:2208.05090). arXiv. <https://doi.org/10.48550/arXiv.2208.05090>

SLM

Hugging face:

Models list: <https://huggingface.co/models>

Documentation: <https://huggingface.co/docs>

Also courses that walk you through parts of the documentation:

<https://huggingface.co/learn>

Relevant papers:

Z. Yu, S. Liu, P. Denny, A. Bergen, and M. Liut, "Integrating Small Language Models with Retrieval-Augmented Generation in Computing Education: Key Takeaways, Setup, and Practical Insights," in Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1, in SIGCSETS 2025. New York, NY, USA: Association for Computing Machinery, Feb. 2025, pp. 1302–1308. doi: 10.1145/3641554.3701844.

S. Liu, R. Ye, C. Eaton, B. Simion, and M. Liut, "A Comparative Study of Student Perspectives on Technical Writing Feedback Quality: Evaluating LLMs, SLMs, and Humans in Computer Science Topics," in Artificial Intelligence in Education, E. G. Blanchard, G. Chen, M. Chi, and S. Isotani, Eds., Cham: Springer Nature Switzerland, 2027, pp. 577–592. doi: 10.1007/978-3-032-29760-0_57.

S. Liu, Z. Yu, F. Huang, Y. Bulbulia, A. Bergen, and M. Liut, “Can Small Language Models With Retrieval-Augmented Generation Replace Large Language Models When Learning Computer Science?,” in Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1, in ITiCSE 2024. New York, NY, USA: Association for Computing Machinery, Jul. 2024, pp. 388–393. doi: 10.1145/3649217.3653554.

Datasets

- <https://coauthor.stanford.edu/>
- Can also find some at: <https://www.kaggle.com/search>
- Someone we could reach out to and see if we can get this data: <https://tools-competition.org/winner/college-writing-dataset/#>

Other approaches and considerations

(This is the approach used during my undergraduate degree. It’s very different from the approach we’ll use, but I hope it gives some idea on the types of technical writing we want to target) L. Zhang et al., “Embedding and Scaling Writing Instruction Across First- and Second-Year Computer Science Courses,” in Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1, in SIGCSE 2023. New York, NY, USA: Association for Computing Machinery, Mar. 2023, pp. 610–616. doi: 10.1145/3545945.3569729.

A. Willis, P. Charlton, and T. Hirst, “Developing Students’ Written Communication Skills with Jupyter Notebooks,” in Proceedings of the 51st ACM Technical Symposium on Computer Science Education, in SIGCSE ’20. New York, NY, USA: Association for Computing Machinery, Feb. 2020, pp. 1089–1095. doi: 10.1145/3328778.3366927.

Y. Liu, W. Xiong, Y. Xiong, and Y. B. Wu, “Generating timely individualized feedback to support student learning of conceptual knowledge in Writing-To-Learn activities,” J. Comput. Educ., vol. 11, no. 2, pp. 367–399, Jun. 2024, doi: 10.1007/s40692-023-00261-3.

A. M. Renner, “Computational Considerations in Correcting User-Language”.

Writing Background

This section is more relevant for me (Amber) than for you, but if you're curious this is some of the background of writing in CS. Please note I haven't read all of these, so some may not be relevant.

M. Allen, "An In-Depth Exploration of Undergraduate Computer Science Students' Perspectives on Learning Technical Writing in Community," *ACM Trans. Comput. Educ.*, vol. 26, no. 2, p. 23:1-23:28, Feb. 2026, doi: 10.1145/3785357.

Alpizar-Chacon, "Excited, Skeptical, or Worried? A Multi-Institutional Study of Student Views on Generative AI in Computing Education." Accessed: Aug. 28, 2026. [Online]. Available: <https://dl.acm.org/doi/epdf/10.1145/3769994.3770008>

C. B. Cilliers, "Student perception of academic writing skills activities in a traditional programming course," *Computers & Education*, vol. 58, no. 4, pp. 1028–1041, May 2012, doi: 10.1016/j.compedu.2011.12.001.

R. Engineer, N. Sibia, M. Kaler, B. Simion, and L. Zhang, "Early Computer Science Students' Perspectives Towards The Importance Of Writing," in *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, in ITiCSE 2024. New York, NY, USA: Association for Computing Machinery, Jul. 2024, pp. 332–338. doi: 10.1145/3649217.3653576.

S. B. Hopton and S. Zurhellen, "Exploring Writing Transfer in an Advanced Curriculum for Technical and Professional Writing and Computer Science Majors," in *2022 IEEE International Professional Communication Conference (ProComm)*, Jul. 2022, pp. 192–198. doi: 10.1109/ProComm53155.2022.00038.

N. Hynninen, "Impact of digital tools on the research writing process: A case study of collaborative writing in computer science," *Discourse, Context & Media*, vol. 24, pp. 16–23, Aug. 2018, doi: 10.1016/j.dcm.2018.01.005.